Lecture 26 — Predict a system through time

Fri Oct 23

Last updated: October 8, 2026.

Boyd video (optional): Lecture 26 — linear dynamical systems · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Lab 4, Lachniet Ch. 2 (matrix operations), and Lachniet's introductory programming/loops material. Related objectives: Lectures. Schedule.

Same day: Quiz 7 (Week 8 systems / RREF — **not** dynamics) and Lab 4.

Bring paper. After the quiz: short lecture on implementing $x_{t+1} = Ax_t$, then Lab 4.

Learning objectives

By the end of class, students should be able to:

1. Implement a linear dynamical model.
2. Write or complete a short iterative calculation.
3. Validate the first steps by hand.
4. Compare results under altered parameters or initial states.
5. Record an unexplained long-run pattern for later investigation.

Warm-up

Let

$$A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 10 \\ 0 \end{bmatrix}.$$

Hand-compute $x_1 = Ax_0$ and $x_2 = Ax_1$. These two checks are exactly what you will use in lab to validate a loop.

Fill in

Implement $x_{t+1} = Ax_t$ in software:

- Enter the matrix A and the initial state x_0.
- In a loop: replace x by _____ (Octave `A*x`, Python `A @ x`).
- Optionally store each x_t in a list / array for plotting.

Validate. Before trusting 50 steps, check that the code's x_1 and x_2 match your _____ calculation.

Compare. Change one parameter in A, or change x_0, re-run, and describe the difference in words.

Mystery log. When a long-run pattern surprises you, write it down — we return to it in Week 13 (eigenvectors), not today.

Worked in class

W1. Complete the iteration. Fill the blanks so this produces $x_0, \ldots, x_5$ (either language is fine on the board):

```
x = x0
for t = 1 to 5:
    x = ______
    print / store x
```

W2. Hand-check then compare. With $A = \text{diag}(1.2,\ 0.5)$ and $x_0 = (5, 4)$:

(a) Compute x_1 by hand.
(b) If software reports $x_1 = (6, 2)$, is the code consistent?
(c) What-if: replace x_0 by $(5, 0)$. After many steps, which component dominates?

Your turn

Y1. Pseudocode check. Someone writes `for t = 1 to 10: x = A*x` but never prints the initial x_0. How many states after x_0 does the loop produce? What should they add if they also want x_0 recorded?

Y2. Validate. $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, $x_0 = (1, 0)$. Hand-compute x_1 and x_2. If a plot of the first component reads $1, 0, -1, 0, 1, ...$, does that match?

Y3. Mystery sentence. In one sentence, record a long-run question from Lecture 25 or today's warm-up that you cannot yet explain with a formula (save for Week 13).

Lab 4, same day

After the short lecture: Lab 4 — Predict a System Through Time. Implement $x_{t+1} = Ax_t$, iterate, hand-check the first steps, compare parameters, and record an unexplained long-run pattern. Preview the one-pager; write on the hard copy in class.

Reminder: Quiz 7 today covers Week 8 (systems / RREF), not this dynamics lab.

Answers (Your turn)

Y1. The loop produces $x_1, \ldots, x_{10}$ (ten states after x_0). Print or store x once *before* the loop to include x_0.

Y2. $x_1 = (0, -1)$, $x_2 = (-1, 0)$. First components: 1, 0, -1, ... — yes, matches the start of that plot sequence.

Y3. Sample: "Why does diag$(1.1, 0.5)$ grow for some initial states and decay for others?" (Any clear unexplained pattern from class is fine.)